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Short-term efficacy and safety of sitagliptin treatment in long-term stable renal recipients with new-onset diabetes after transplantation

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ABSTRACT

Background. New-onset diabetes after transplantation (NODAT) is a common complication after renal transplantation. There are limited available oral drugs to treat hyperglycaemia in this population owing to reduced renal function, potential interactions with immunosuppressive drugs and adverse effects such as hypoglycaemic events that may increase the cardiovascular risk. This study was initiated to investigate efficacy and safety of sitagliptin treatment that may represent a novel alternative in renal transplant recipients.

Methods. Nineteen long-term stable renal transplant recipients with NODAT were included in a controlled, cross-over

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study and randomized to first receive either sitagliptin 50–100 mg/day or a sitagliptin-free period of 4 weeks. Median age (interquartile range, IQR) was 67 (62–72) years (12 males/7 females), all studied 1 (1–3) year after transplantation. The immunosuppressive regimen was a triple calcineurin inhibitor-based therapy. Oral glucose tolerance test (OGTT) with insulin and C-peptide responses and laser Doppler (LD) flowmetry assessment of endothelial function were performed at baseline and after each treatment period. Home measurements of plasma glucose were performed daily during the study.

Results. The median (IQR) first- and second-phase insulin secretion responses increased significantly by 56.3% (45.2–112.6%, P = 0.005) and 39.3% (26.5–81.0%, P = 0.006), respectively, following sitagliptin treatment as compared with no sitagliptin treatment. Fasting and 2-h plasma glucose concentrations fell significantly {0.9 mmol/L [0.5–1.7 mmol/L (16.2 mg/dL), P = 0.003] and 2.9 mmol/L [0.5–6.4 mmol/L (52.3 mg/dL), P = 0.004], respectively}, as did also home measurements of plasma glucose. Endothelial function and plasma markers of cardiovascular risk were unaffected. No serious adverse events were observed. Two mild and asymptomatic hypoglycaemic episodes were observed in combination with glipizide.

Conclusions. Sitagliptin increases insulin secretion and reduces fasting and postprandial plasma glucose in renal transplant recipients with NODAT. The short-term treatment was well tolerated, and sitagliptin seems safe in this population.

Keywords: insulin, NODAT, renal transplantation, sitagliptin

INTRODUCTION

New-onset diabetes after transplantation (NODAT) develops in 10–15% of all renal transplant recipients within 10 weeks after renal transplantation [1] and has been associated with increased risk of cardiovascular disease and impaired patient survival [2–6]. Current guidelines for treating type 2 diabetes mellitus (T2DM) primarily recommend life style interventions and, if needed, additional treatment with oral glucose-lowering agents or insulin [7]. Drug options are limited in renal transplant recipients owing to reduced renal function and potential pharmacokinetic interactions with immunosuppressive drugs [8–10].

Sitagliptin (Januvia®, Merck Sharp & Dohme Ltd) is an oral active, potent and selective dipeptidylpeptidase-4 (DPP-4) inhibitor, which inhibits the degradation of the incretin hormones glucagon-like peptide-1 (GLP-1) and glucose inhibitory peptide (GIP). This increases pancreatic insulin secretion and decreases glucagon release, and since these effects are glucose dependent, sitagliptin does not induce hypoglycaemia or weight gain [11–13]. A recent *in vitro* research also indicates that sitagliptin may reduce inflammatory responses [14]. This would be of special potential interest in renal transplant recipients since systemic markers of inflammation are increased in these patients [15] and also associated with adverse cardiovascular effects [16]. The elimination of sitagliptin is primarily via renal excretion and only marginally affected by CYP metabolism (CYP3A and CYP2C8) [17]. Dose adjustments are recommended for patients with moderate or severe renal impairment [18].

The primary objective of the present study was to investigate the effect of 4 weeks sitagliptin treatment on insulin secretion in renal transplant recipients with NODAT. The impact on plasma glucose, insulin sensitivity, endothelial function and safety parameters such as calcineurin inhibitor/everolimus levels and changes in renal function were investigated as secondary objectives. In addition, the effect of cardiovascular risk on plasma inflammatory markers was studied.

SUBJECTS AND METHODS

Patients

The Laboratory for Renal Physiology, Oslo University Hospital, Rikshospitalet, performs standard clinical follow-up of renal transplant recipients in Norway both 10 weeks and 12 months after transplantation. The examinations include assessment of glucose metabolism status by an oral glucose tolerance test (OGTT), and the results are stored in a database. Potential participants in the present study were identified by screening for patients with fasting plasma glucose of ≥7.0 mmol/L (126 mg/dL) or 2-h plasma glucose of ≥11.1 mmol/L (200 mg/dL) post-glucose load (diagnosis of NODAT according to the WHO criteria [19]) in the database. Adult renal transplant recipients with NODAT were eligible for inclusion if they had been transplanted more than 1 year ago, had stable renal function (<20% variation in S-creatinine during the last 2 months) and had stable maintenance prednisolone dose for the last 3 months. Also patients who were on oral antidiabetic treatment were eligible for inclusion as long as they continued with these drugs in unchanged doses. Exclusion criteria consisted of severe liver disease, estimated GFR (eGFR) of <25 mL/min/1.73 m^2, pregnant/nursing mothers and patients on insulin treatment. The recruitment took place over a period of 4 months where the patients received oral and written information. Each patient gave written informed consent before they were consecutively included in the study.

Study design

The study was a prospective, open, randomized, controlled, cross-over study. Patients were randomized to receive either sitagliptin once daily for 4 weeks followed by 4 weeks with no sitagliptin treatment or *vice versa*. The randomization was performed by a premade list of treatment sequence, and the consecutively included patients were assigned the next number on the list, which was not available to the physicians recruiting the patients (AH and TJ). The sitagliptin dose was adjusted to renal function. Patients with eGFR of ≥50 mL/min/1.73 m^2 received 100 mg/day, whereas patients with eGFR of 25–49 mL/min/1.73 m^2 received 50 mg/day. All concomitant therapy was held stable during the study period. The participants were in addition informed to keep a stable life style, including diet and exercise.

Oral glucose tolerance test (OGTT) and LD flowmetry for determination of endothelial function were performed at baseline and after each treatment period (Figure 1). Each patient

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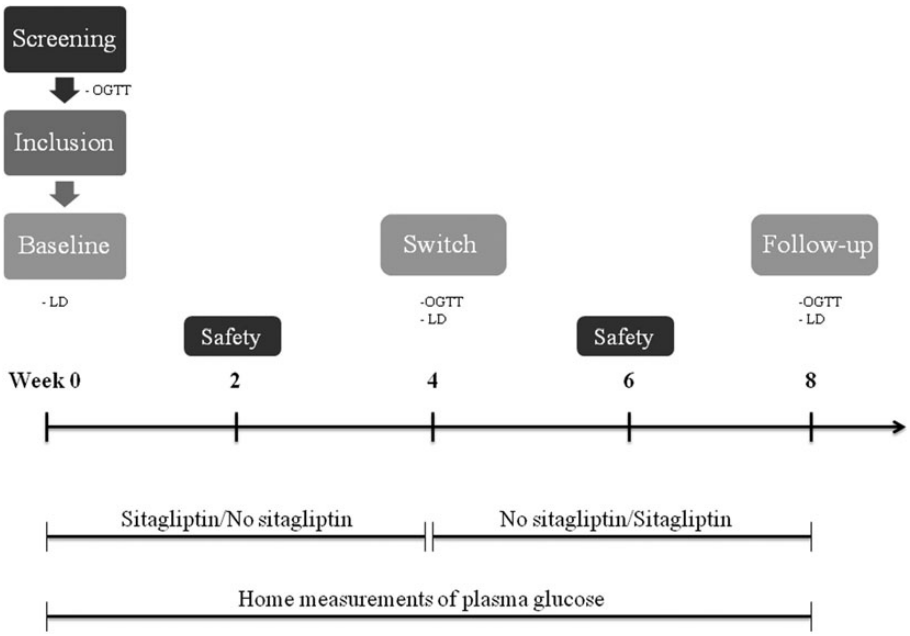

FIGURE 1: Study schedule. OGTT, oral glucose tolerance test; LD, laser Doppler flowmetry.

also self-monitored plasma glucose concentrations (Freestyle Lite, Abbot®) four times every day (fasting and 2 h after breakfast, lunch and dinner) during the study.

For safety reasons, patients were examined 2 weeks into each treatment period. On these visits, plasma glucose was measured and a clinical evaluation and standard safety analyses including trough calcineurin inhibitor/everolimus blood concentrations were performed (Figure 1). Sitagliptin was provided in blister packages, and compliance was controlled by tablet counting after the sitagliptin treatment period.

The study was performed according to the Declaration of Helsinki and was approved by the Regional committee for Medical Research Ethics and by the Medicines Control Agency, Norway (EudraCT number: 2008-001442-45). The study was registered on ClinicalTrials.gov (NCT00740363, 21 August 2008).

Oral glucose tolerance test and glucose measurements

On each experimental day, the patients presented in the morning for conduction of OGTT after an overnight (10 h) fast. Sitagliptin and all concomitant therapies were withheld and administered after completion of the blood sampling. Blood samples for measurement of plasma glucose, serum insulin and C-peptide were drawn before (0 min) and 30 and 120 min after administration of an oral glucose load of 75 g. Classification of glucose tolerance was performed using standard criteria [19].

Analytical methods

Plasma glucose was measured from whole-blood samples using Hemocue® Glucose 201 Analyser (Ängelholm, Sweden). Serum insulin was analysed with an Enzyme-Linked Immunosorbent Assay (ELISA) kit (Human Insulin, Invitrogen Ltd, KAQ1251, Paisley, UK). Insulin and all other ELISA analyses were performed in two parallel samples. Serum C-peptide was measured using an ELISA kit (Dako, K6220, Glostrup, Denmark). Plasma markers related to cardiovascular risk were also analysed with ELISA kits (Abcam plc, Cambridge, UK) including micro-C-reactive protein (μCRP), soluble tumour necrosis factor receptor-1 (sTNFR-1), vascular cell adhesion molecule-1 (VCAM-1), adiponectin and interleukin (IL)-10 (detection limit, <5 pg/mL).

Standard blood samples for analysis of biochemical parameters and trough concentrations of the immunosuppressive drugs were analysed by routine clinical methods for the hospital central laboratory.

Microvascular endothelial function

Endothelial function was validated by LD flowmetry of the forearm arterioles after stimulation by transdermal iontophoresis of acetylcholine (ACh) and after a post-occlusive reactive hyperaemia (Rh) test as previously described [20]. The effect parameter used was the area under the flux versus time curve (AUC), following the fourth iontophoretic stimulation (AUC_{Ach}) and after the Rh-test (AUC_{Rh}), respectively.

Calculations

The first- and second-phase insulin secretions were calculated from the equations published by Stumvoll *et al.* [21]:

$$\text{Secr1. phase} = 1283 + 1.829 \times \text{Ins}_{30} - 138.7 \times \text{Gluc}_{30} + 3.772 \times \text{Ins}_{0},$$

$$\text{Secr2.phase} = 287 + 0.4164 \times \text{Ins}_{30} - 26.07 \times \text{Gluc}_{30} + 0.9226 \times \text{Ins}_{0},$$

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where Ins_0 and Ins_{30} (pmol/L) are serum insulin concentrations at 0 and 30 min after an OGTT, respectively, and $Gluc_{30}$ (mmol/L) represents plasma glucose 30-min post-glucose load.

The homeostasis model assessment (HOMA) was also used to calculate the insulin release index ($Secr_{HOMA}$) as follows [22]:

$$Secr_{HOMA} = Ins_0 \times 3.33/(Gluc_0 - 3.5),$$

where $Gluc_0$ (mmol/L) represents fasting plasma glucose.

C-peptide was corrected for glucose and renal function (C-peptide/glucose creatinine ratio; CPGCR) by using the following formula [23]:

$$CPGCR = (C\text{ - peptide} \times 100)/(\text{glucose} \times \text{creatinine}).$$

All calculations were performed using measures for C-peptide (nmol/L), glucose (mmol/L) and serum creatinine (μmol/L) obtained from the same fasting blood sample for each time point.

Insulin sensitivity index (ISI) modified for transplant recipients (tx) was calculated according to the following equation [24]:

$$ISI_{TX} = 0.208 - 0.0032 \times BMI - 0.0000645 \times Ins_{120} - 0.00375 \times Gluc_{120},$$

where Ins_{120} (pmol/L) and $Gluc_{120}$ (mmol/L) are insulin and glucose concentrations 2-h post-glucose load, respectively.

Estimated GFR was calculated using the modification of diet in renal disease formula [25].

Safety

The participants were subjected to a physical examination and were specifically interviewed for adverse experiences at each study visit. The participants also logged any hypoglycaemic episodes in their logbooks throughout the study (severe hypoglycaemia: plasma glucose <2.0 mmol/L; unconsciousness or the need of assistance, moderate hypoglycaemia: plasma glucose between 2.0 and 3.9 mmol/L). Safety laboratory analyses as described were done every second week.

Statistical considerations

Number of patients. Ten patients were needed to be included in the study to obtain a power of 80% ($\alpha = 0.05$) to detect a 25 ± 20% increase in insulin secretion after sitagliptin treatment as compared with control (no sitagliptin treatment) [26, 27]. Only six patients had, however, provided complete evaluable results after a pilot study; hence, a total of twenty-five patients who early after transplantation had been shown to have NODAT based on OGTT were included. Upon inclusion in the study, six patients did not meet the criteria for NODAT but had impaired glucose tolerance only. Thus, these patients were excluded from the statistical analyses but had similar responses to sitagliptin treatment as the remaining 19 patients with NODAT (data not shown).

Analysis plan. All metabolic parameters were analysed as difference in change at the end of each treatment period from baseline: 'sitagliptin minus baseline' versus 'no sitagliptin minus baseline'. The Wilcoxon signed-rank test was used for comparing paired data, and a P-value of <0.05 was considered statistically significant. The results are given as median (IQR) if not otherwise explained. All statistical analyses were performed using SPSS Statistics for Windows (Version 17.0, Chicago: SPSS, Inc.).

RESULTS

Patients

Baseline characteristics of the 19 Caucasian renal transplant recipients are given in Table 1. Twelve patients had drug-naïve NODAT whereas seven patients were long-term-treated for their NODAT with glipizide ($n = 5$) or glimepiride ($n = 2$) at the time of inclusion. The immunosuppressive treatment consisted of cyclosporine A ($n = 13$), tacrolimus ($n = 4$) or everolimus ($n = 3$, one patient with combined cyclosporine and everolimus treatment) in combination with prednisolone ($n = 18$, mean dose: 5.1 ± 0.9 mg/day) and/or mycophenolate ($n = 17$). All patients were negative for hepatitis C virus and antibodies. Five patients had cytomegalovirus infection prior to NODAT diagnosis whereas five patients had autosomal dominant polycystic kidney disease.

Insulin secretion and sensitivity

Figure 2 shows the difference in change in insulin secretion against baseline. The median (IQR) first- and second-phase insulin secretion responses increased significantly by 56.3% (45.2–112.6%, P = 0.005) and 39.3% (26.5–81.0%, P = 0.006), respectively, following sitagliptin treatment as compared with control ($n = 14$).

Following 4 weeks without sitagliptin treatment, median (IQR) response in CPGCR was 0.18 (0.16–0.23), and with sitagliptin treatment, it was significantly higher: 0.22 (0.18–0.29) (C-peptide/glucose × creatinine) (P = 0.001, $n = 18$). This difference represented a median (IQR) relative CPGCR increment of 27.1% (8.2–40.7%).

The $Secr_{HOMA}$ data were consistent with our main results with a median (IQR) increment of 45.3% (13.4–76.1%) following sitagliptin treatment as compared with control (P = 0.015, $n = 15$).

The insulin sensitivity was also significantly different between the two treatment periods, with a median (IQR) of 25.3% (1.6–49.9%, P = 0.04, $n = 15$) higher sensitivity after sitagliptin treatment.

Table 1. Demographic data at baseline given as median (IQR)(in numbers)

Age (years)	67 (62–72)
Gender (male/female)	12/7
BMI (kg/m^2)	28.4 (25.2–29.8)
Time since transplantation (years)	1 (1–3)
Donor (LD/DD)	5/14
Blood pressure (mmHg):	
Systolic	147 (131–160)
Diastolic	86 (76–90)
Renal function:	
Estimated GFR (mL/min/1.73 m^2)	61 (43–85)
Sitagliptin dose (50 mg/100 mg)	2/17

LD, living donor; DD, deceased donor.

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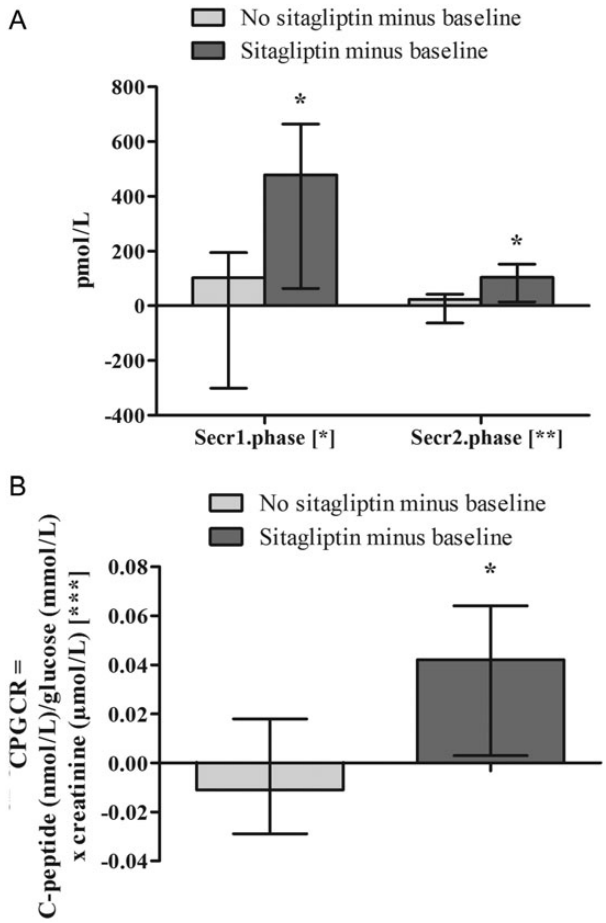

FIGURE 2: Median (IQR) change in insulin secretion from baseline for (**A**) first- and second-phase insulin secretion ($n = 14$), and (**B**) C-peptide values corrected for glucose and creatinine (CPGCR, $n = 18$). Asterisks indicate statistically significant differences between the two treatment periods: [*] represents $1283 + 1.829 \times Ins_{30} - 138.7 \times Gluc_{30} + 3.772 \times Ins_0$; [**] represents $287 + 0.4164 \times Ins_{30} - 26.07 \times Gluc_{30} + 0.9226 \times Ins_0$[21]; [***] represents C-peptide/glucose × creatinine [23].

Glucose metabolism

Figure 3 shows the effect of sitagliptin treatment on glucose metabolism. The median (IQR) fasting plasma glucose was significantly lower after 4 weeks of sitagliptin treatment: 8.0 (7.1–8.3) mmol/L (144 mg/dL) versus 7.0 (5.9–7.6) mmol/L (126 mg/dL) (P = 0.003, $n = 19$). Similar results in plasma glucose were also seen for the 30-min glucose concentrations during the OGTT: [11.8 (10.7–13.5) mmol/L (212.6 mg/dL) versus 10.8 (9.5–12.1) mmol/L (194.6 mg/dL), P = 0.007, $n = 17$] and for the 120-min concentrations [14.3 (11.4–16.4) mmol/L (257.7 mg/dL) versus 9.7 (8.9–11.4) mmol/L (174.8 mg/dL), P = 0.004, $n = 17$].

Sitagliptin treatment also resulted in significant and consistent reductions in home measured plasma glucose throughout the day. The median (IQR) reductions were as follows: 0.9 (0.3–1.2) mmol/L (16.2 mg/dL) fasting, before breakfast (P = 0.003, $n = 18$), 0.4 (0.0–1.9) mmol/L (7.2 mg/dL) 2 h after breakfast (P = 0.006, $n = 17$), 0.5 (0.3–2.5) mmol/L (9.0 mg/dL) 2 h after lunch (P = 0.009, $n = 13$) and 0.5 (0.0–1.2) mmol/L (9.0 mg/dL) 2 h after dinner (P = 0.002, $n = 16$).

BMI and blood pressure

Sitagliptin had no significant effects on median (IQR) BMI [0.0 (−2.7–0.3) kg/m^2, P = 0.48], systolic [−4 (−14–1) mmHg, P = 0.13] or diastolic [−2 (−6–5) mmHg, P = 0.38] blood pressure.

Endothelial function

Sitagliptin had no significant effect on endothelial function. The relative median (IQR) difference in AUC_{ACh} and AUC_{Rh} was −52.1 AU × min (−154.7–213.1 AU × min, P = 0.65, $n = 17$) and −6.5 AU × min (−16.0–4.5 AU × min, P = 0.38, $n = 16$), respectively, as compared with control measurements.

Plasma markers

Sitagliptin did not induce any significant changes in the selected plasma markers of cardiovascular risk: µCRP, VCAM-1, adiponectin and sTNFR-1. Median (IQR) changes were as follows: −368 (−1710–873) ng/mL (P = 0.41, $n = 17$), 1.3 (−69–74) ng/mL (P = 0.98, $n = 18$), 3421 (−9438–29205) ng/mL (P = 0.23, $n = 18$) and −13 (−35–4) pg/mL (P = 0.20, $n = 18$), respectively. IL-10 concentrations were below the lower limit of detection (<5 pg/mL) in all samples.

Safety

Sitagliptin was well tolerated, and the compliance was high (99.6%). Only two patients reported a single missed intake of sitagliptin. One patient was withdrawn from the study due to night sweating. The symptoms were not related to low plasma glucose concentrations, but they disappeared after treatment withdrawal and recurred with reintroduction of sitagliptin. None of the other study participants reported symptoms that could be related to sitagliptin, including nausea. Asymptomatic moderate hypoglycaemia was recorded in two patients, each at one single occasion. Both patients received glipizide in addition to sitagliptin.

A significant increase in median (IQR) uric acid of 25 (−2–45) µmol/L, P = 0.02, and minor changes in alkaline phosphatase of −3.0 (−8.5–1.0) U/L, P = 0.07; GT of −2.5 (−12.3–0.3) U/L, P = 0.005; and ALAT of −1.0 (−4.3–0.0) U/L, P = 0.04 were seen after sitagliptin treatment. No clinically relevant changes were observed in any other safety parameters (Table 2). No significant changes were shown in trough concentrations of the immunosuppressive drugs following sitagliptin treatment: cyclosporine A (P = 0.45), tacrolimus (P = 0.07), everolimus (P = 1.00) and mycophenolate (P = 0.65).

DISCUSSION

The major findings of the present study were that sitagliptin significantly increased first- and second-phase insulin secretion and reduced fasting and postprandial plasma glucose concentrations and that sitagliptin was well tolerated in renal transplant recipients. Home measurements of plasma glucose showed reductions in consistence with OGTT. These findings are in accordance with those obtained with sitagliptin in non-

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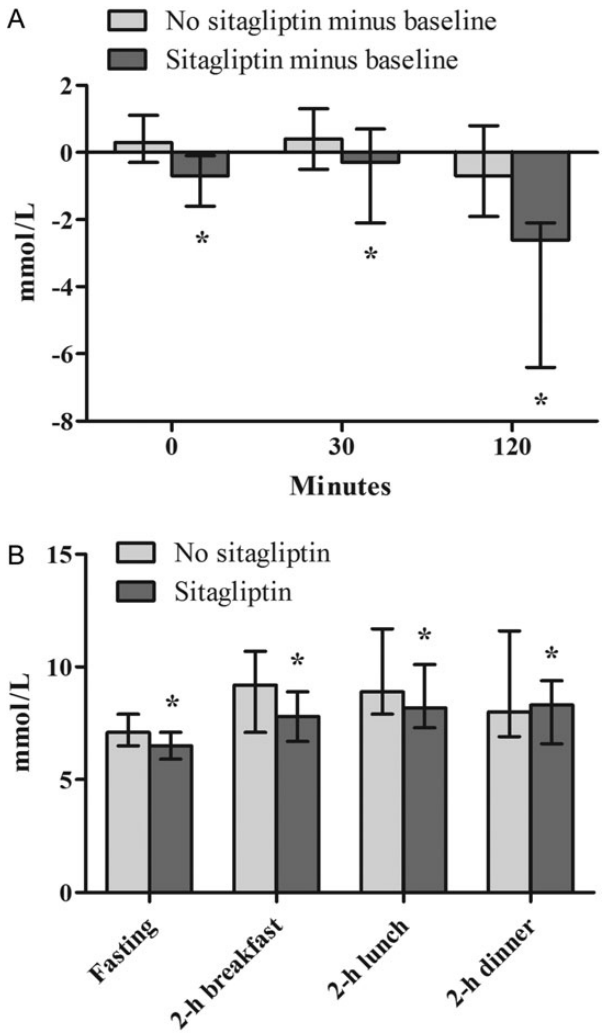

FIGURE 3: Median (IQR) effect on glucose metabolism following (**A**) OGTT, given as change from baseline, and (**B**) home measurements of plasma glucose, given as median absolute concentrations of the last 14 days in each treatment period. Asterisks indicate statistically significant differences between the two treatment periods. 2-h breakfast, 2-h lunch and 2-h dinner denote 2 h after breakfast, 2 h after lunch and 2 h after dinner, respectively.

transplanted T2DM patients inadequately controlled with metformin [28].

C-peptide responses were also studied as a secondary confirmatory measure of the sitagliptin-associated insulin response. Since renal transplant recipients have impaired renal function, the C-peptide values corrected for creatinine and glucose (CPGCR) were used [23]. The significant increase in CPGCR of 27.1% further supports that sitagliptin improved β-cell function in these patients as has previously been shown in other cohorts [29, 30]. Evidence suggests that impaired insulin secretion rather than insulin resistance seems to be the main contributing factor for NODAT development [31, 32]. DPP-4 inhibitors may, therefore, be especially beneficial in NODAT patients, since they improve insulin response without exhausting the beta cell, as opposed to for example sulphonylureas [33]. Other oral drugs such as metformin are also available. However, metformin is generally not used in renal transplant recipients in our clinic due to their reduced renal function and hence increased risk for metabolic acidosis, although it has been suggested by others [33, 34]. Impaired insulin secretion has been proposed to be an even bigger problem in renal transplant recipients than that in non-transplanted patients with T2DM or milder forms of disturbed glucose metabolism [35], underlining the novelty of the findings in the present study.

A general treatment goal for patients with T2DM is to lower HbA1c to 7% or below, as this has been shown to reduce diabetic microvascular complications [36]. Since patients with NODAT often have advanced atherosclerosis and high cardiovascular risk, the reduction in HbA1c should not be at the cost of hypoglycaemic episodes or weight gain, as this may increase their overall cardiovascular risk [37]. In the present study, sitagliptin was only used for 4 weeks, which obviously is too short to fully evaluate its effect on HbA1c. Such an effect has been shown in NODAT patients, where a reduction in HbA1c from 7.2 to 6.7% was found after 3 months of sitagliptin treatment [38]. A 3-month treatment with vildagliptin has also been shown to be of potential benefit in stable renal transplant recipients with IGT [39].

We could not find any significant effect on endothelial function with sitagliptin treatment. In a previous study in renal transplant recipients with IGT/NODAT, we found that rosiglitazone treatment of 4 weeks was associated with a significant improvement in microvascular endothelial function [40]. The present results, therefore, suggest that the previous positive effect on endothelial function could be due to pleiotropic effects of rosiglitazone and not its glucose-lowering effect *per se*. We have previously also found a significant association between endothelial dysfunction and insulin resistance in renal transplant recipients [41]. This was speculated partly to be due to impaired circulation of the peripheral skeletal musculature. The fact that sitagliptin treatment improved insulin sensitivity but not endothelial function in the present study could apparently contradict these results. However, it is important to bear in mind that assessment of microvascular endothelial function using LD flowmetry has some limitations as the mean day-to-day coefficient of variation in response to ACh is high (34.4–42.0%) [42]. Since sitagliptin has favourable effects on insulin secretion, insulin sensitivity might have improved in part through a decrease in glucose toxicity [43].

C-reactive protein is involved in atherosclerosis [44]. Increased expression of VCAM-1 and hypoadiponectinaemia have been associated with impaired endothelial function [45, 46], whereas a low level of sTNFR-1 has been associated with improved renal outcomes in patients with diabetic nephropathy [47]. Only minor, non-significant changes in these plasma markers were shown in the present study.

Sitagliptin was well tolerated, which is in agreement with studies conducted in patients with T2DM [48, 49]. Of particular relevance for transplant recipients, sitagliptin seems to be safe also considering potential interactions with immunosuppressive treatment. The present data, where no significant changes in trough concentrations of calcineurin inhibitors/everolimus were shown, substantiate that there are no previous

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Table 2. Biochemical parameters, given in median (IQR)

	Baseline	No sitagliptin	Sitagliptin	P-value[a]
Haemoglobin (g/dL)	13.8 (12–15)	14.0 (12–15)	13.9 (12–15)	0.93
Thrombocytes (10^9/L)	242 (216–260)	230 (208–279)	226 (206–264)	0.14
Leucocytes (10^9/L)	7.1 (5.9–7.9)	6.5 (5.7–8.0)	7.3 (6.3–8.7)	0.65
Uric acid (μmol/L)	406 (315–453)	385 (333–452)	421 (355–488)	0.02
Sodium (mmol/L)	141 (140–143)	142 (141–143)	142 (141–144)	0.22
Potassium (mmol/L)	3.7 (3.6–3.9)	3.8 (3.5–4.2)	3.8 (3.7–4.0)	0.46
Calcium (mmol/L)	2.4 (2.3–2.4)	2.3 (2.2–2.5)	2.3 (2.3–2.4)	0.78
Phosphate (mmol/L)	0.9 (0.9–1.1)	1.0 (0.9–1.1)	1.0 (1.0–1.2)	0.27
Bilirubin total (μmol/L)	10.0 (6.0–11.5)	8.0 (6.0–10.5)	8.0 (6.0–10.0)	1.00
ASAT (U/L)	23.0 (20–27)	22.0 (20–26)	21.0 (19–27)	0.31
ALAT (U/L)	23.0 (18–26)	24.0 (18–27)	23.0 (16–26)	0.04
LD (U/L)	217 (185–234)	207 (186–240)	192 (178–226)	0.15
Alkaline phosphatase (U/L)	73.5 (66–95)	78.0 (62–83)	70.0 (62–82)	0.07
GT (U/L)	33.0 (23–51)	41.0 (22–54)	34.0 (20–44)	0.005
Urea (mmol/L)	8.4 (6.3–11.2)	8.2 (6.4–12.2)	8.7 (7.0–12.2)	0.68
Creatinine (μmol/L)	105 (78–135)	103 (85–131)	109 (83–140)	0.20
eGFR (mL/min/1.73 m^2)	61.0 (43–85)	60.0 (45–74)	58.0 (41–75)	0.11
HbA1c (%)	6.9 (6.7–7.3)	6.7 (6.5–7.2)	6.7 (6.3–7.3)	0.01
CRP (mg/L)	2.1 (1.3–2.6)	1.7 (1.1–3.0)	1.6 (1.3–2.6)	0.49

[a]Sitagliptin minus baseline versus no sitagliptin minus baseline.

reports that show any relevant drug–drug interactions with sitagliptin [50].

The significant decrease in alkaline phosphatase (accompanied by decreased GT) and increased uric acid levels in the present study were also found in a recent 12-week sitagliptin study in non-transplanted T2DM-patients, suggesting that these actions are due to sitagliptin and its activation of incretin hormones [51].

Clinical trials should ideally be controlled, randomized, and preferably also blinded in order to avoid bias as far as possible. A limitation of the present study was that it was not blinded or placebo-controlled. Regression towards the mean was reduced, however, due to randomization of which study period the participants started with (sitagliptin treatment versus no sitagliptin treatment). A majority of the participants were prescribed cyclosporine treatment rather than tacrolimus, which to some degree limits the results since tacrolimus is well recognized to be more diabetogenic [52]. The relatively short follow-up time in this study also to some degree limits the results in comparison with the studies in which longer treatment periods were applied. DPP-4 inhibitors such as sitagliptin have advantages in the context of moderate-to-severe renal dysfunction. However, since the present study does not provide safety data in renal transplant recipients with eGFR of <25 mL/min/1.73 m^2, our data cannot be extended to treatment of patients in this population.

CONCLUSION

A 4-week treatment with sitagliptin significantly increased insulin secretion and sensitivity and reduced fasting and postprandial plasma glucose. The drug was also well tolerated in renal transplant recipients. Thus, sitagliptin represents a novel treatment alternative for renal transplant recipients with NODAT, but long-term data on such therapy are warranted.

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CONFLICT OF INTEREST STATEMENT

The results presented in this paper have not been published previously in whole or part, except in abstract format.

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